

App_Flow.md

Application Flow Document

Project: Bloom

Document: Application Flow & User Journey

Version: 1.0

Status: Product Flow Specification

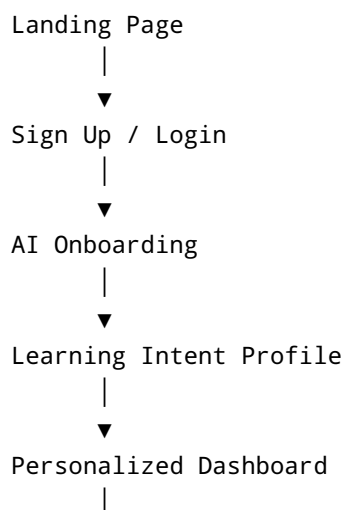
1. Purpose

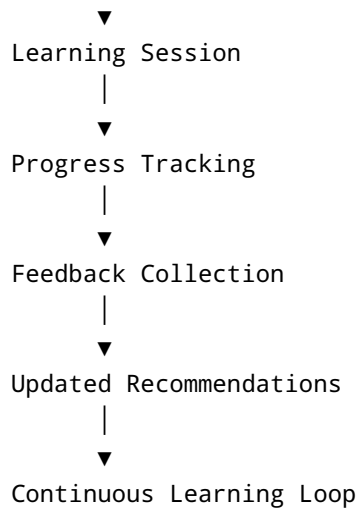
This document defines how users interact with Bloom from the moment they discover the product until they become long-term learners.

Unlike the PRD, which defines features, this document defines **how those features connect together into a seamless experience.**

The objective is to ensure every user action has a clear path, predictable behavior, and measurable outcome.

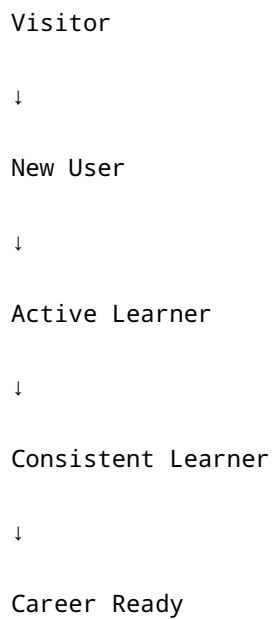
2. High-Level Application Flow





3. User Lifecycle

Bloom supports five stages of a learner's journey.



Each stage unlocks progressively richer personalization.

4. First-Time User Flow

Step 1 — Landing Page

User arrives from:

- Search Engine
- YouTube
- Referral
- Social Media
- University
- Direct Link

Actions:

- Learn about Bloom
 - View product overview
 - Install Extension / Open Web App
 - Create Account
-

Step 2 — Authentication

Supported methods:

- Google
- GitHub
- Email & Password

After successful authentication:

↓

Proceed to onboarding.

5. AI Onboarding Flow

Purpose:

Understand the learner before making recommendations.

Flow:

Welcome

↓

Career Goal

↓

Current Experience

↓

Known Skills

↓

Preferred Learning Style

↓

Daily Learning Time

↓

Learning Objectives

↓

Generate Learning Profile

Output:

Learning Intent Profile

6. Dashboard Flow

After onboarding:

User enters the Personalized Dashboard.

Sections:

- Today's Learning
- Continue Learning

- Recommended Resources
- Skill Progress
- Learning Streak
- Upcoming Revision
- Career Progress
- Notifications

Primary Call-to-Action:

Start Learning Session

7. Learning Session Flow

Start Session

↓

Learning Objective

↓

Recommended Resource

↓

Consume Resource

↓

Practice Activity

↓

Reflection

↓

Progress Update

↓

Next Recommendation

A learning session should feel like a guided experience rather than a list of links.

8. Recommendation Flow

Every recommendation follows the same pipeline.

Learner Opens Dashboard

↓

System Reads Learning Profile

↓

Identify Skill Gap

↓

Retrieve Candidate Resources

↓

Rank Resources

↓

Generate Explanation

↓

Display Recommendation

Each recommendation includes:

- Why this resource?
- Estimated duration
- Difficulty
- Skills covered
- Expected outcome

9. Progress Tracking Flow

Whenever a learner completes an activity:

Resource Completed

↓

Update Learning History

↓

Estimate Skill Mastery

↓

Update Progress

↓

Recalculate Roadmap

↓

Generate Next Recommendation

Progress should update automatically without manual intervention.

10. Search Flow

The learner may search manually.

Search Query

↓

Intent Detection

↓

Resource Retrieval

↓

Ranking

↓

Results

↓

Save / Learn

Supported searches:

- Skill
- Career
- Creator
- Topic
- Technology
- Project

11. Guard Mode Flow

Purpose:

Reduce distractions.

Learning Session Starts

↓

Detect Distractions

↓

Prompt User

↓

Hide/Reduce Distracting Content (where supported)

↓

Resume Learning

Guard Mode should always remain optional and transparent.

12. Revision Flow

Bloom periodically recommends revision.

Knowledge Decay Prediction

↓

Identify Weak Concepts

↓

Generate Revision Queue

↓

Revision Session

↓

Update Mastery

Revision timing is determined by Bloom's learning intelligence.

13. Feedback Flow

Feedback can be explicit or implicit.

Explicit

- Like
- Dislike
- Save
- Report

Implicit

- Watch completion
- Session duration
- Resource abandonment
- Repeated searches

Flow:

Feedback

↓

Feedback Processing

↓

Update Learner Profile

↓

Improve Future Recommendations

14. Career Roadmap Flow

Users can inspect their roadmap at any time.

Career Goal

↓

Required Skills

↓

Completed Skills

↓

Current Skill

↓

Upcoming Skills

↓

Career Readiness

Users may:

- Change goals
- Reorder roadmap
- Add custom skills

15. Notification Flow

Possible notifications:

- Daily reminder
- Revision reminder
- Learning streak
- Milestone achieved
- New recommendation
- Roadmap updated

Flow:

Event

↓

Notification Engine

↓

Priority Evaluation

↓

Delivery

↓

User Action

Notifications should encourage consistency without becoming intrusive.

16. Settings Flow

Users can configure:

- Career goals
- Learning preferences
- Daily study time
- Notification settings
- Privacy options
- Connected accounts
- Guard Mode

Changes immediately update the learner profile.

17. Error Handling Flow

Example:

Recommendation service unavailable.

Recommendation Request

↓

Service Failure

↓

Fallback Recommendation

↓

Display Friendly Message

↓

Retry in Background

The learner should always have a usable experience even when parts of the system are unavailable.

18. Returning User Flow

Login

↓

Continue Previous Session

↓

Resume Roadmap

↓

Recommended Next Step

↓

Learn

↓

Track Progress

Bloom minimizes friction by allowing users to resume where they left off.

19. Long-Term Learning Loop

Learn

↓

Practice

↓

Complete

↓

Feedback

↓

Profile Update

↓

Better Recommendations

↓

Learn Again

This continuous cycle enables increasingly personalized learning over time.

20. Administrative Flow (Future)

Administrator capabilities may include:

- View platform analytics
- Manage content metadata
- Monitor recommendation quality
- Configure system settings
- Review reported resources
- Manage institutional users

Administrative tools are isolated from learner workflows.

21. Edge Cases

The application should gracefully handle:

- New users with no history
- Learners changing career goals
- Missing external resources
- Offline periods (future)
- Authentication failures
- Duplicate accounts
- Recommendation service degradation

Each scenario should provide a meaningful fallback path.

22. State Transitions

A learner moves through application states as follows:

Unauthenticated

↓

Authenticated

↓

Onboarded

↓

Active Learning

↓

Session Complete

↓

Profile Updated

↓

Recommendation Refreshed

↓

Next Session

Each transition should be atomic and recoverable to avoid inconsistent user states.

23. User Journey Summary

Journey	Outcome
Visitor → Sign Up	New account created
Sign Up → Onboarding	Learning profile generated

Journey	Outcome
Onboarding → Dashboard	Personalized experience begins
Dashboard → Learning Session	Active learning starts
Learning Session → Progress	Skills updated
Progress → Feedback	AI receives learning signals
Feedback → Recommendations	Future recommendations improve
Continuous Learning	Long-term skill development

24. Application Principles

Every application flow should adhere to the following principles:

- Minimize unnecessary steps.
- Personalize from the first session.
- Explain important AI decisions.
- Keep learners focused on progress rather than content consumption.
- Recover gracefully from failures.
- Preserve user control over recommendations and settings.
- Continuously adapt based on learner interactions.

App Flow Summary

Bloom is designed around a continuous learning cycle rather than isolated interactions. Every screen, workflow, and transition contributes to a single objective:

Guide learners from intention to mastery through clear, personalized, and adaptive learning experiences.

The application flow emphasizes low friction, transparent AI assistance, and continuous improvement, ensuring that each interaction strengthens the learner's profile and improves future recommendations.